# Enes Altun

Web: altunenes.github.io

## CONTACT


- enesaltun2@gmail.com
- github.com/altunenes
- personal site
- ORCID
- Google Scholar
- LinkedIn


## SKILLS

**AI & ML**

- ML/LLM deployment on local systems
- Multimedia:Fast multimodal data processings via GStreamer
- Parallel data processing: Apache Arrow, DataFusion

**Graphics**

- Graphics programming: wgpu, wgsl/glsl

**Domain Expertise**

- Cognitive Science, ML, EEG signal processing, Psychophysics

## EDUCATION

2020

**Akdeniz University, Department of Psychology**
B.A. in Psychology — Antalya, Turkey

2022

**Akdeniz University, Department of Psychology**
M.A. in Psychology — Antalya, Turkey

- **Thesis:** *Visual processing of different-race and great-ape faces in the context of spatial frequency: an ERP study.* Visual Cognition (2024)

## RESEARCH EXPERIENCE

2017 — 2021

**Undergraduate Research Assistant**
Akdeniz University Cognitive Neuroscience Lab — Antalya, Turkey

- Designed and conducted psychophysics experiments using EEG and signal processing.
- Built small electronic devices (GSR sensors, robotic fingers) for psychophysical experiments.

2021 — 2023

**Graduate Research Assistant**
Akdeniz University Cognitive Neuroscience Lab — Antalya, Turkey

- Image and signal processing, human and computer vision, machine learning.

## INDUSTRY EXPERIENCE

Jul 2023 — Present

**Rust Developer**
Empatik AI — Stockholm, Sweden

- Built a cross platform, GPU accelerated multimodal desktop app.
- Worked on high perf multimedia and on-device AI in Rust + ONNX Runtime for efficient local inference.
- Optimized GPU & CPU utilisation across multiple ML/LLM models to minimise latency and memory overhead.
- Developed real time parallel audio/video pipelines with GStreamer that integrate with AI models.

## SELECTED PUBLICATIONS & POSTERS

**Altun, E.**, Er, M. N., & Gülbetekin, E. (2024). The early visual processing of faces in a basic classification task: an ERP study of spatial frequency, other-race and other-species effect. *Visual Cognition*, 1–24. doi.org/10.1080/13506285.2024.2415721

**Altun, E.**, Er, M. N., & Gülbetekin, E. *(Under review)*. Illusory faces evoke a face-selective N170 from fine, but not coarse, spatial frequencies: an ERP study of face pareidolia. Earlier version: doi.org/10.2139/ssrn.4341900

Gülbetekin, E., **Altun, E.**, Er, M. N., Fidancı, A., Keskin, P., & Steenken, D. (2021). Effects of right or left face stimulation on self and other perception in enfacement illusion. *Social Neuroscience*, 16(2), 189–205. doi.org/10.1080/17470919.2021.1886983

Gülbetekin, E., Bayraktar, S., Kantar Gül, D., Varlık Özsoy, E., Er, M. N., **Altun, E.**, & Fidancı, A. (2023). Does tactile stimulation of the face affect the processing of other faces? Neural and behavioural effects of facial touch. *Social Neuroscience*. doi.org/10.1080/17470919.2023.2245126

Gülbetekin, E., Fidancı, A., **Altun, E.**, Er, M. N., & Gürcan, E. (2023). Effects of mask use and other-race on face perception, emotion recognition, and social distancing during the COVID-19 pandemic. *Asian Journal of Social Psychology*, 1–16. doi.org/10.1111/ajsp.12570

Gülbetekin, E., Fidancı, A., **Altun, E.** (2023). How do we process faces from different ethnic groups and species? *Society for Neuroscience Annual Meeting*, Washington, DC.

Gülbetekin, E., **Altun, E.**, Er, M. N. (2020). Does own-species matter? Holistic face perception for great apes. *International Behavioral Neuroscience Society (IBNS) Online Poster Sessions*, Texas, USA, 3–7 August 2020, pp. 1–1.

**Altun, E.**, Er, N., Gülbetekin, E. (2019). Does genetic similarity matter? Face-recognition performance for primates. *Fechner Day 2019: 35th Annual Meeting of the International Society for Psychophysics*, Antalya, Turkey, pp. 91–92.

Er, N., **Altun, E.**, Fidancı, A., et al. (2019). Effect of stimulated face side on multisensory perception. *Fechner Day 2019*, Antalya, Turkey, pp. 79–79.

**Altun, E.** (2023). altunenes/scramblery: 2.1.1. *Zenodo*. doi.org/10.5281/zenodo.7484576

## SOFTWARE PROJECTS

2026 — **Willcaster** — Rust · Bevy

Solo-developed strategic roguelike dungeon crawler for Steam. The player draws paths for an autonomous companion whose trust and desires shape whether it obeys or defies.

2024 — **Cuneus** — Rust · WGSL

GPU compute engine for creating and shipping WGSL shader programs. Built with wgpu and GStreamer for real-time video/audio processing; multi pass pipelines and hot shader reload.

2025 — **parakeet-rs** — Rust

Very fast ASR library wrapping NVIDIA's Parakeet models via ONNX Runtime. Real time streaming and speaker diarization.

2022 — **Scramblery** — Rust

Cross platform GPU accelerated desktop app for advanced image and video scrambling (Fourier phase, diffeomorphic warping with optical flow). Facial detection and efficient frame processing via GStreamer.

2023 — **Butter2D** — Rust

Pure Rust implementation of the Butterworth filter for spatial frequency filtering of images, with an online WASM demo.

2023 — **Weber–Fechner** — Rust

Template for psychophysics experiments implementing the Weber–Fechner law of perception, using the Bevy engine with WASM support.